

# Zainab Oyekade

Edinburg, Texas 78541 • [zainaboyekade@gmail.com](mailto:zainaboyekade@gmail.com) • 956-360-6696

• Github: <https://github.com/zainaboyekade-dot> • Portfolio: <https://zainaboyekade-dot.github.io>

## OBJECTIVE

Computer Engineering undergraduate specializing in embedded systems, firmware development, and intelligent software. Passionate about designing reliable hardware-software solutions through low-level programming, system architecture, and applied engineering projects.

## EDUCATION

**University of Texas Rio Grande Valley**

**B.S. Computer Engineering | GPA: 4.0 | Expected 2029**

Relevant Coursework: Programming Fundamentals (C++), Digital Systems

Honors: President's List

## SKILLS

**Coding Languages:** C++, C, Python, SQL, JavaScript, HTML/CSS

**Libraries/Tools:** STM32CubeIDE, STM32 HAL, clang-tidy, Streamlit, Pandas, Plotly, Jupyter Notebook, Git/GitHub, Arduino IDE, Blynk

**Concepts:** Embedded Systems, Firmware Development, Microcontrollers, Hardware Interfaces (I2C/SPI/UART), Sensor Integration, Static Analysis, System Design, Debugging, Data Analysis

**Soft Skills:** Problem Solving, Communication, Teamwork, Adaptability, Fast Learning

## EXPERIENCE & PROGRAMS

**UTRGV Regional Security Operations Center (RSOC) | InfoSec Student Analyst**

**June 2026 - Present**

- Complete structured RSOC training and job shadowing to learn security operations procedures, analyst workflows, and information security practices.
- Assist security analysts with reviewing alerts, analyzing event details, and documenting findings through SOC ticket workflows.
- Support monitoring activities using SIEM tools while gaining hands-on experience with security event analysis and incident response procedures.

**Global Career Accelerator | Data Analytics Projects (SQL & Python)**

**Jan 2026 - April 2026**

- Developed SQL queries using joins, CTEs, and aggregate functions to analyze Intel sustainability and device lifecycle datasets.
- Performed Python-based data analysis using Pandas and Plotly to evaluate web traffic metrics and user engagement trends.
- Analyzed sustainability metrics including energy savings, CO<sub>2</sub> reduction, and regional performance patterns to generate data-driven insights.

**Telstra Virtual Experience (Forage) | Software Engineering**

**May 2025 - Jan 2026**

- Developed solutions in a simulated software engineering environment focused on APIs, system design, and workflows
- Applied structured engineering practices to solve system design and implementation challenges

## PROJECTS

**AstraRT | STM32 Embedded Firmware Runtime Framework**

**July 2026**

C, STM32F446RE, STM32 HAL, I2C, SPI, UART, GPIO, BME280, BMI160

- Designed and developed a modular STM32F446RE embedded runtime framework in C, implementing hardware abstraction, sensor integration, and application-level task management.
- Built a layered firmware architecture with custom abstraction interfaces above STM32 HAL, organizing peripheral drivers, runtime services, and system components for reusable embedded development.
- Implemented communication interfaces using 4 hardware communication interfaces (I2C, SPI, UART, and GPIO), integrating BME280 and BMI160 sensor drivers for structured hardware data acquisition and monitoring.
- Developed runtime services including time-based task scheduling, telemetry, command handling, and system diagnostics, validated through real hardware testing and runtime demonstrations.

**CodeLens-X | Hybrid C/C++ Static Analysis Tool with AI Explanations**

**April 2026**

Python, OpenAI API, clang-tidy, Streamlit

- Built a hybrid C/C++ static analysis pipeline integrating clang-tidy diagnostics, Python parsing, and LLM-powered

explanations to transform compiler warnings into structured debugging reports.

- Designed a Streamlit-based interface for interactive code analysis and visualization of issues
- Automated issue extraction using regex-based parsing of compiler output into structured data objects

#### **Digital Twin-Based Robotic Bending System | Research Project**

**July 2026 - August 2026**

Python, JavaScript, HTML/CSS, Data Visualization, API Integration

- Developed a full-stack dashboard interface for a digital twin-based robotic bending system, incorporating 4 visualization components for monitoring springback predictions and manufacturing process data.
- Built frontend visualization features and backend data workflows to integrate model outputs into an interactive monitoring environment.
- Implemented prediction history tracking and live monitoring capabilities to support comparison of bending results and process behavior.
- Designed dashboard architecture to support future integration of sensor and vision-based data from the physical robotic system.

#### **Smart Plant Watering System | Embedded Systems Project**

**Oct 2025 - Dec 2025**

C++, Arduino, Blynk, Soil Moisture Sensor, Relay Module

- Designed and developed an automated irrigation system integrating soil moisture sensing, relay-based pump control, and IoT monitoring capabilities.
- Programmed microcontroller logic to process sensor readings and trigger watering actions based on real-time moisture thresholds.
- Integrated Blynk for wireless monitoring and remote system control through a connected microcontroller platform.

#### **CERTIFICATIONS**

- **Global Career Accelerator** - Python and Data Certification
- **Global Career Accelerator** - Querying Data Certification
- **Global Career Accelerator** - AI Professional Skills Certification
- **Global Career Accelerator** - Intercultural Skills Certification